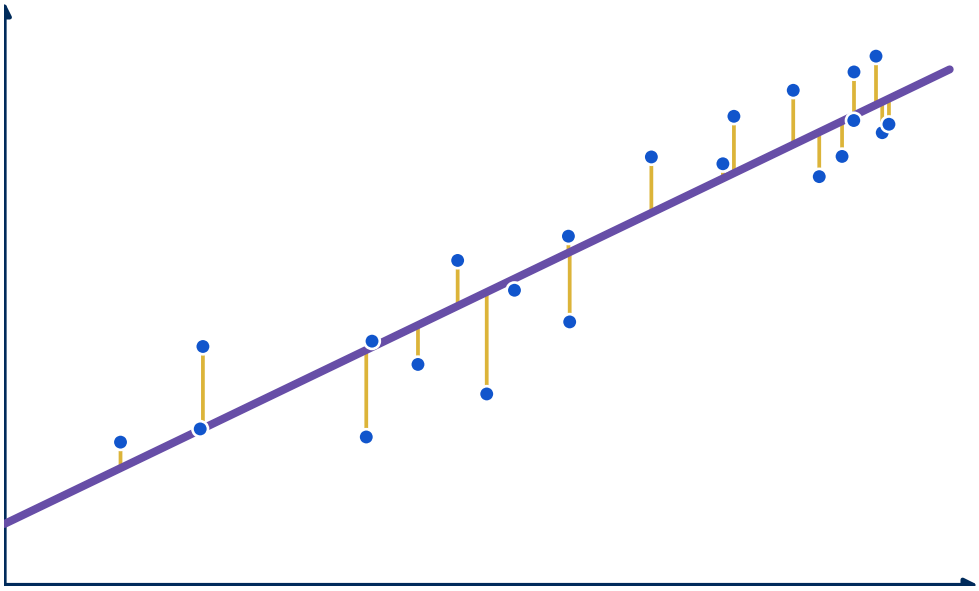


UNIT 01 · CHAPTER 4

# Linear Regression

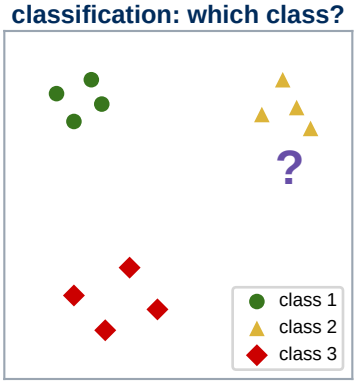
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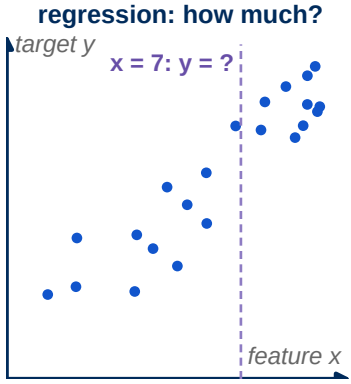
William Theisen · University of Notre Dame

# Classification and regression

Our **previous models**, **k-Nearest Neighbors** and **Decision Trees**, were built for **classification**.

A **classification task** is one in which we try to assign a _____ to a sample.




In **regression tasks**, we try and assign a _____ to each sample.

**Regression** is another form of _____.

**Quick check (old exam question):** is a regression task supervised or unsupervised learning?

# Linear regression

A **linear regression** is a technique to fit a line to our data, allowing us to make predictions.

## Regression vocabulary

Unfortunately, because linear regression is so old, it **has a lot of vocab specific to it**:

[4.1] a line with one feature

$$\hat{y} = w_1 x_1 + w_0$$

- $\hat{y}$: _____
- $w_1$, $w_0$: _____
- $x_1$: _____

# Matrix notation

We can represent basically everything as **matrices** and **matrix operations**.

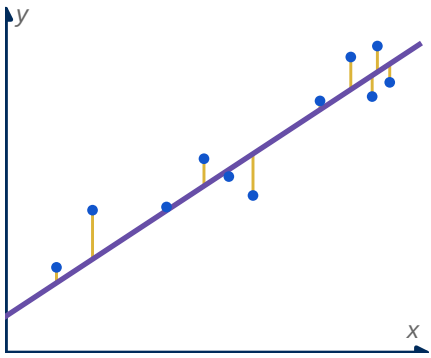
# Multiple features

It's rare that we have datasets with only a single feature. The **iris dataset**, a toy dataset, has 4!

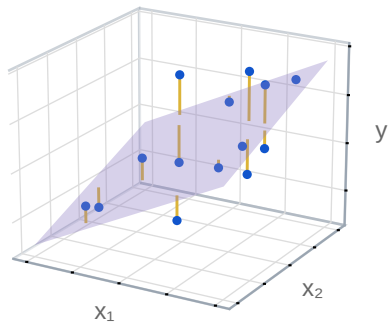
When we perform a **linear regression on** a dataset with **more than one feature**, we call this a _____.

## Hyperplanes

**1 feature: a line**




**2 features: a plane**

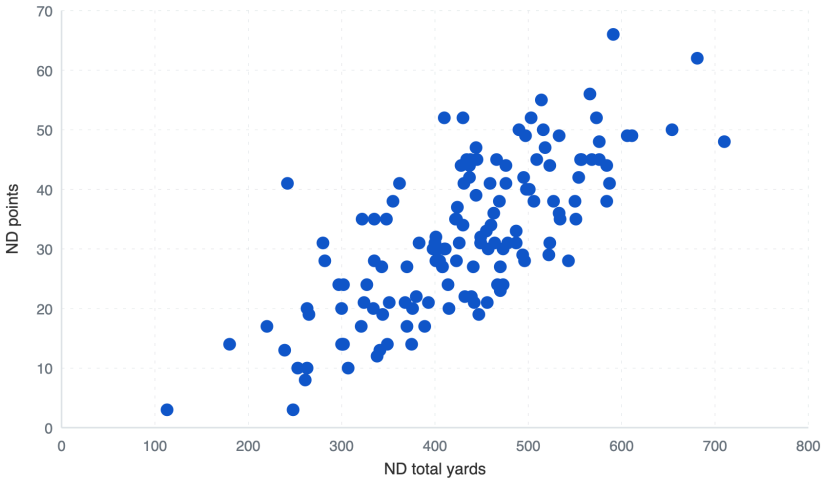



With one feature the model is a line, with two it's a _____ and with more it's a _____.

**Quick check (old exam question):** what would the equation look like for a multiple linear regression with three features?

# Fitting our line

How do we actually find the line that "fits" the best?




Draw the line you think fits best. Which games is it furthest from?

## Optimization problems

We need some way to **measure how well our model fits** our data. We can call this a _____.

For **linear regression** we often use **SSE** or ____.

sum of squared errors

$$[4.3] \quad SSE(w) = \sum_i (y_i - \hat{y}_i)^2$$

mean squared error

$$[4.4] \quad MSE(w) = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

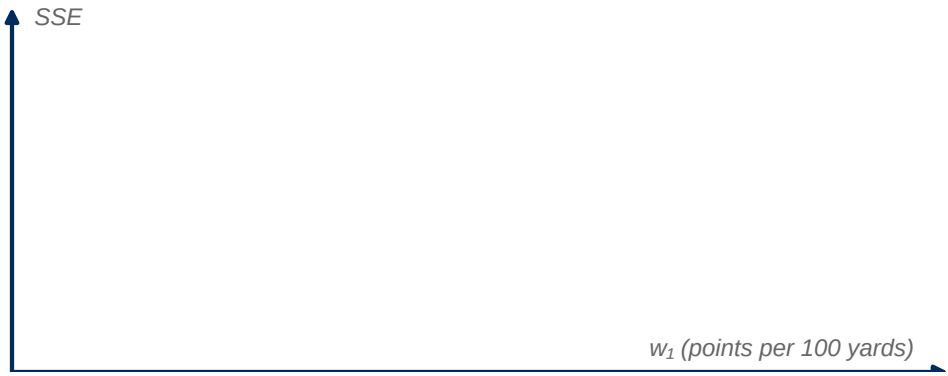
# Finding our weights

So given a **loss function**, our **goal** is to _____ it.

When would a **loss function** be **minimized**? When the _____ is ____.

## Visualizing loss

We can plot the loss function with respect to **w**:



# Ordinary least squares

We can **solve for the gradient equaling 0**:



This is called **Ordinary Least Squares (OLS)**.

# Worked example: OLS by hand

| game | yards | $x_1$ | y (points) |
|---|---|---|---|
| 2019 Michigan | 180 | 2 | 14 |
| 2012 Michigan State | 300 | 3 | 20 |
| 2013 Purdue | 400 | 4 | 31 |
| 2019 Boston College | 501 | 5 | 40 |
| 2017 Temple | 606 | 6 | 49 |

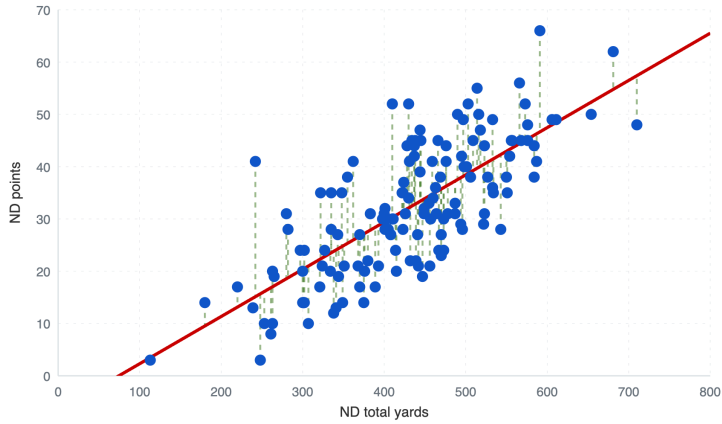
Give $X$ a **column of 1s** for the intercept, then follow the **OLS** recipe.

# Reading the weights

So $\hat{y} = 9.0\,x_1 - 5.2$: about _____.

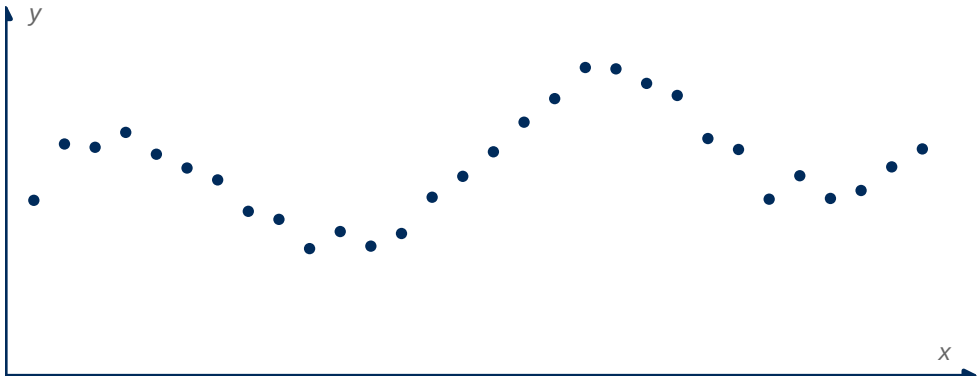
**Quick check:** in 2018 Notre Dame gained 550 yards against Stanford. What does this line predict, and what is the residual if Notre Dame scored 38?

## Linear regression visualization



# Representing curves

How could you use a **linear regression** to model a sine wave?




The model only has to be linear in its _____, not in its inputs.

# A possible issue

The **weight calculation relies on computing the inverse**, which can be **extremely computationally expensive** and **sometimes just literally impossible**.

# An iterative method

Instead of jumping straight to the bottom of the bowl, we can _____ a little at a time.

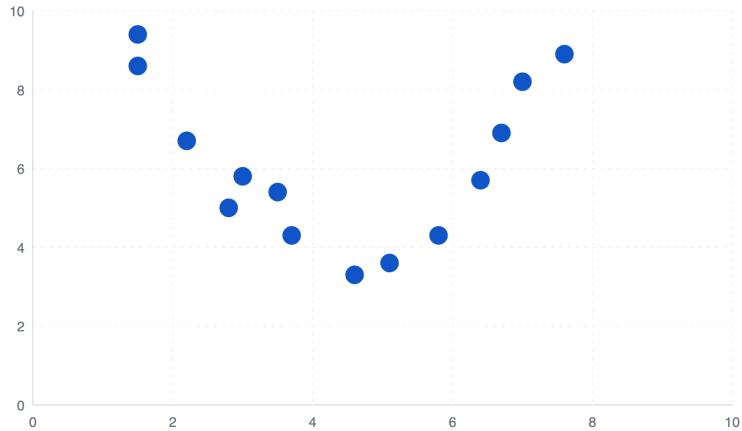


# Maximum likelihood estimation

**Maximum likelihood estimation (MLE)** is a method of estimating the parameters of an _____, **given some observed data**.

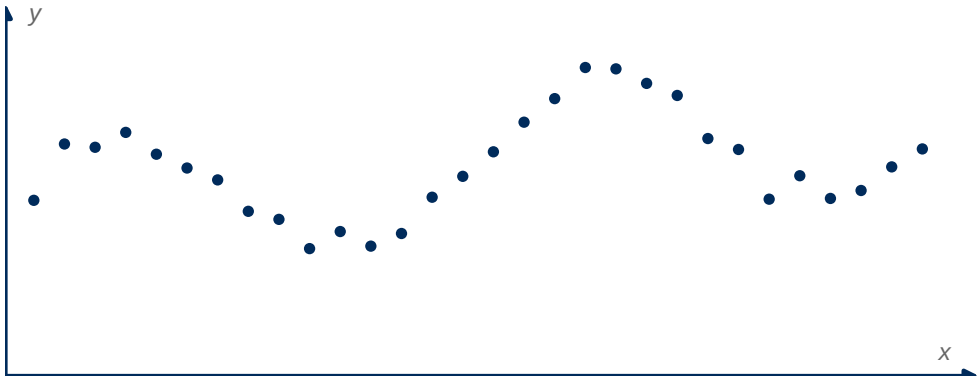
## Overfitting

Unfortunately, **Linear Regressions** (a form of MLE), are **prone to overfitting**.

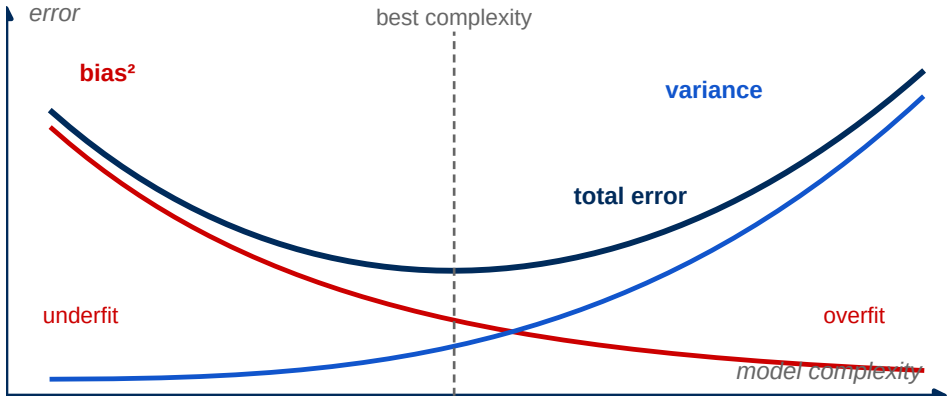


# Underfitting

**Underfitting** is when your model is "too weak" to capture all of the underlying patterns in the data.




## Bias and variance




The **bias–variance tradeoff** describes the **relationship between** a **model's complexity**, the **accuracy** of its predictions, and how well it can make **predictions on unseen data** that were not used to train the model.

# Practice: an old exam question

| $x_1$ | $x_2$ | $y$ |
|---|---|---|
| 1 | 1 | 9 |
| 2 | 2 | 25 |
| 3 | 3 | 50 |
| 4 | 4 | 84 |
| 5 | 5 | 127 |

Given the dataset above, answer the following questions **(Old exam question)**:

**A.** Would a **purely linear model** exhibit **high bias** here or **high variance**? Would it be **overfit** or **underfit**? Explain your reasoning.

**B.** Design a regression equation that would be able to approximate the dataset.

**C.** What role does the **SSE** play in **linear regression**? Why does it matter that our resulting parameter space is convex?

**D.** Look at the $x_1$ and $x_2$ columns. What goes wrong if we fit $\hat{y} = w_0 + w_1 x_1 + w_2 x_2$ with **OLS**?

# Regularization: $L_1$ and $L_2$

To help **prevent overfitting** when using **linear regressions**, we can perform a technique called **regularization**.

The goal of **regularization** is to **encourage models to learn the broader patterns** within the data rather than memorizing it.

## $L_2$ regularization: Ridge

**L2 regularization** (called **ridge regularization**) **encourages** _____ **weights**.

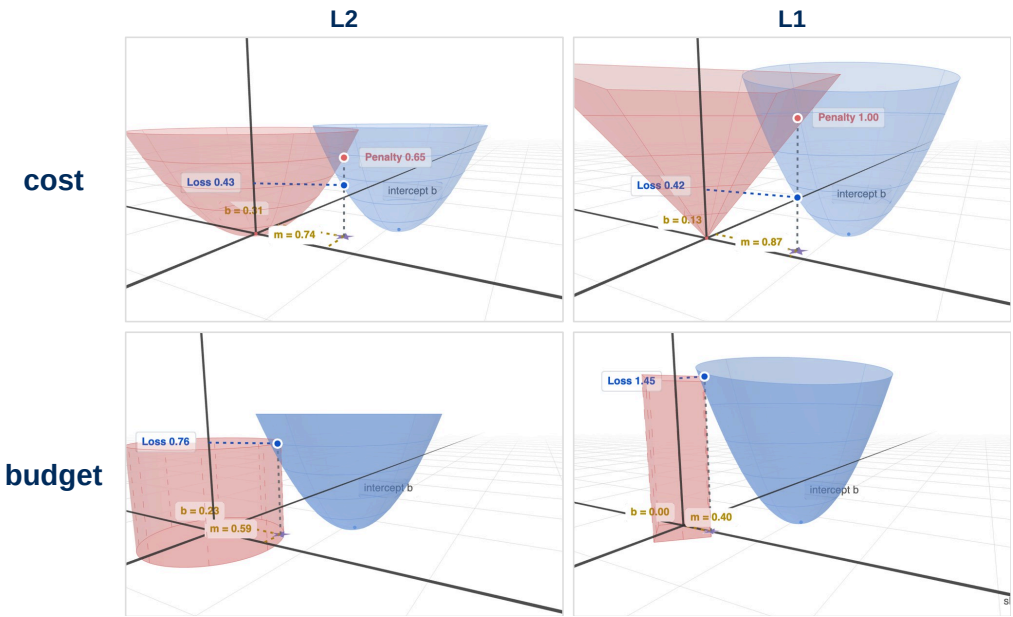
## $L_1$ regularization: LASSO

**L1 regularization** (also called **LASSO**) leads to _____ and **sparse models**.

# Formalizations of regularization

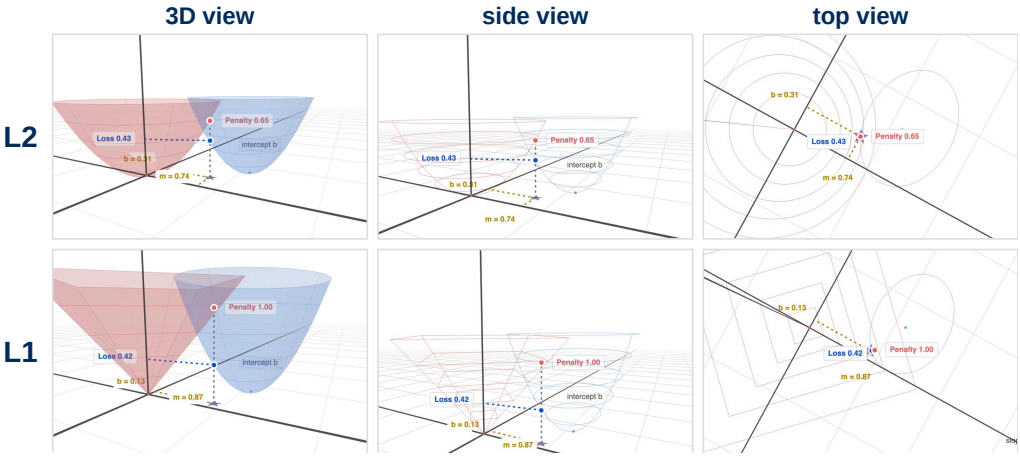
There are two ways to formalize regularization:

- As a _____ placed on the **weights**
- As a _____ for the **weights**



# Regularization as a cost

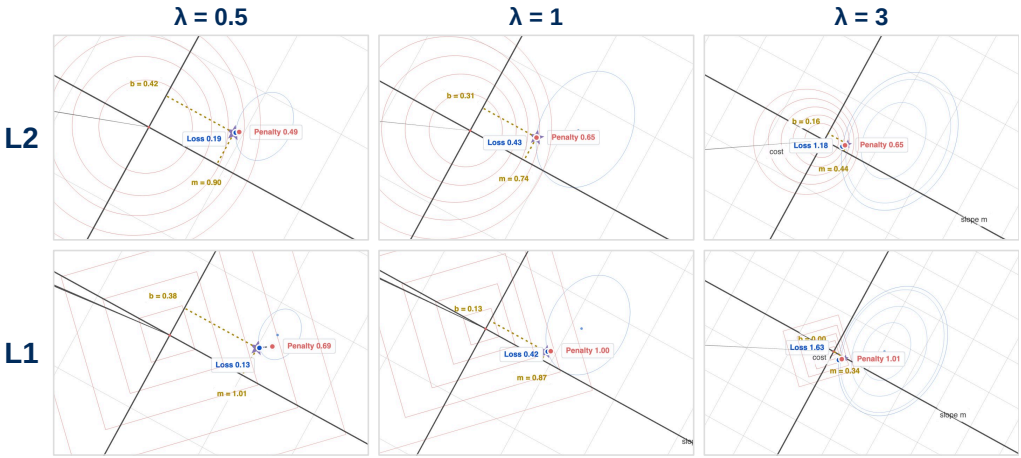
When we formalize **regularization** as a **cost**, we modify the **loss function** such that increasing the weights incurs a "cost".




We use an argument called $\lambda$ to determine how heavily the **cost** is weighed. A larger $\lambda$ increases the **cost**.

# Regularization as a cost, increasing λ

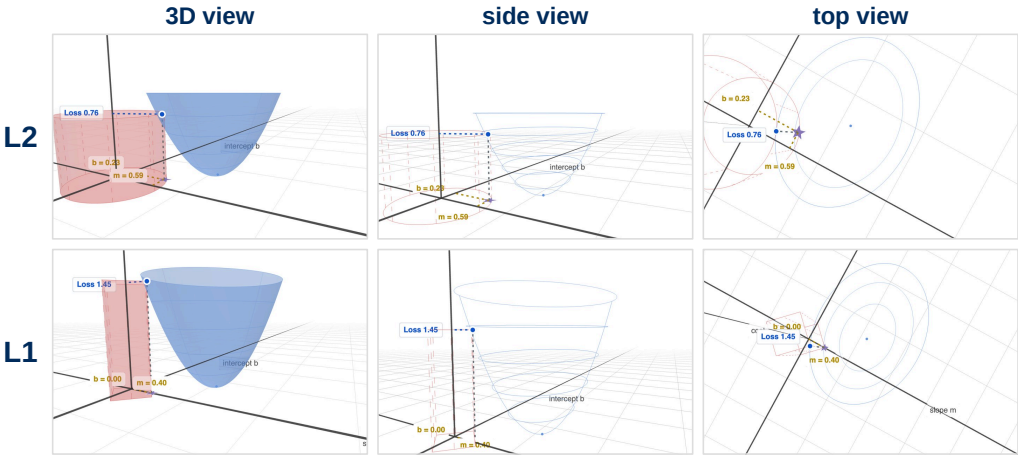
A larger λ increases the **cost** of our weights, typically resulting in the optimum being pulled closer to the origin.




## Regularization as a budget

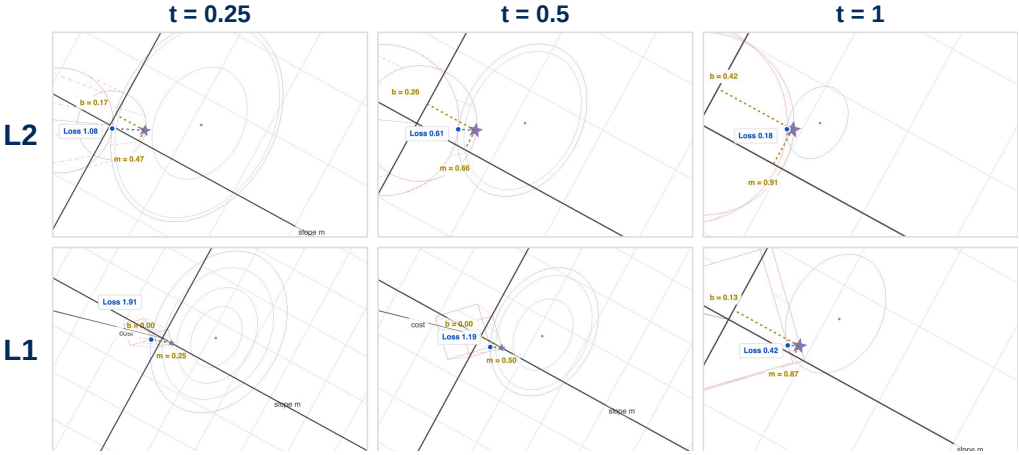
When we formalize **regularization** as a **budget**, we constrain our **loss function** such that the weights chosen must remain within "budget".

# Regularization as a budget




## Increasing $t$

A larger $t$ increases the **budget** for our weights, typically resulting in the optimum being closer to the unconstrained optimum.



# MAP estimation

**L1** and **L2 regularization** are a form of **MAP Estimation**.

**MAP** stands for _____ which is a **Bayesian** approach for **applying a prior to our weights**.

# Review

Complete the review: **L1 regularization** penalizes _____. **L2 regularization** penalizes _____. In a penalty formulation, increasing **lambda** makes the penalty _____. In a budget formulation, increasing the budget makes the constraint _____.

# Matching: regression and regularization

Write the letter of the matching definition next to each term.

| | |
|---|---|
| ___ **Ordinary Least Squares** | **A.** Penalizes the squared size of the weights, encouraging small weights |
| ___ **Maximum Likelihood Estimation** | **B.** Controls how heavily the regularization cost is weighed |
| ___ **MAP Estimation** | **C.** The budget the weights must remain within |
| ___ **L2 (Ridge)** | **D.** The model memorizes noise in the training data instead of the general pattern |
| ___ **L1 (LASSO)** | **E.** The relationship between model complexity, training accuracy, and accuracy on unseen data |
| ___ $\lambda$ | **F.** Solving for the weights by setting the gradient of the loss equal to 0 |
| ___ $t$ | **G.** Estimating the parameters that make the observed data most likely |
| ___ **Underfitting** | **H.** The model is too weak to capture the underlying patterns in the data |
| ___ **Overfitting** | **I.** Penalizes the absolute size of the weights, driving some to exactly 0 |
| ___ **Bias–variance tradeoff** | **J.** A Bayesian approach for applying a prior to our weights |